

Supplement to: What Teachers Should Know about the Bootstrap: Resampling in the Undergraduate Statistics Curriculum

Tim C. Hesterberg
Google
timhesterberg@gmail.com

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This document contains materials that supplement “What Teachers Should Know about the Bootstrap: Resampling in the Undergraduate Statistics Curriculum” (Hesterberg, 2015). There are additional figures, additional discussion of bias, more general discussion of bootstrap confidence intervals, and a description of bootstrap tilting intervals.

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1 Verizon Figures

This section has additional figures related to the Verizon example. Figure 1 has histograms and normal quantile plots for one-sample bootstrap distributions for the ILEC and CLEC means.

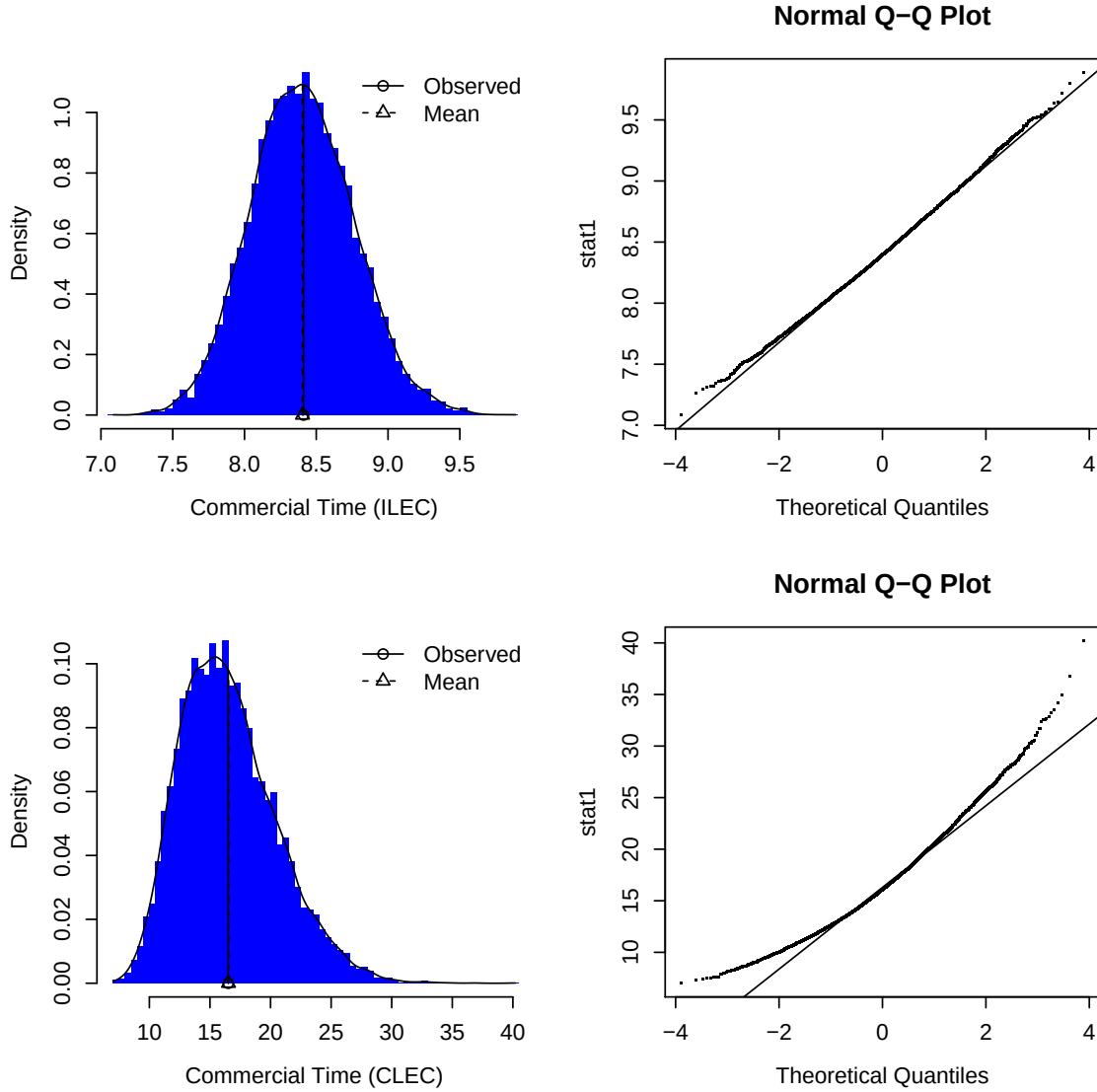


Figure 1: *ILEC and CLEC bootstrap distributions; histogram and normal quantile plots.*

Figure 2 has histograms and normal quantile plots for the two-sample bootstrap and permutation test for the difference in CLEC and ILEC means.

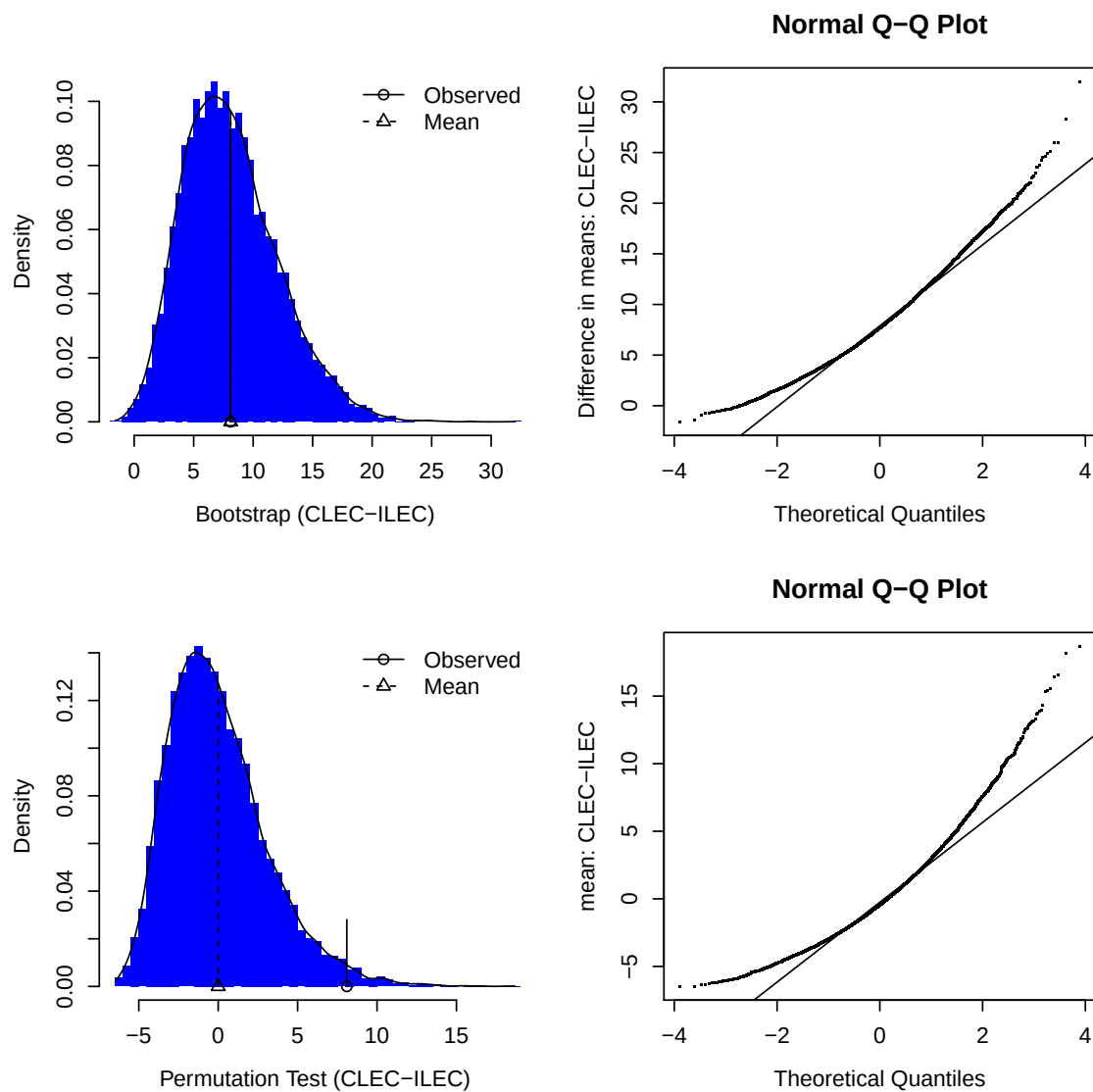


Figure 2: *Two-sample bootstrap and permutation distributions for the difference between CLEC and ILEC means; histogram and normal quantile plots.*

2 Bias

There are subtle issues related to bootstrap bias estimates. The bias estimate derives from the plug-in principle. The bias B of a statistic is

$$B = E(\hat{\theta}) - \theta = E_F(\hat{\theta}) - \theta(F) \quad (1)$$

where E_F indicates sampling from F , and $\theta(F)$ is the parameter for population F . The bootstrap substitutes $\hat{F}$ for F , to give

$$\hat{B} = E_{\hat{F}}(\hat{\theta}^*) - \theta(\hat{F}) = \overline{\hat{\theta}^*} - \hat{\theta}. \quad (2)$$

The bias estimate is the mean of the bootstrap distribution, minus the observed statistic.

For example, we may estimate the bias of $\hat{\sigma}^2 = 1/n \sum (x_i - \bar{x})^2$ as an estimate of σ^2 , or $\hat{\sigma}$ as an estimate of σ . For the CLEC data, the bootstrap gives

	Observed	SE	Mean	Bias
s^2	380.38946	267.201641	362.36533	-18.024129
varhat	363.85079	255.584179	346.61032	-17.240471
sigmahat	19.07487	7.191139	17.17274	-1.902133

$\hat{\sigma}^2 = (22/23)s^2 = 363.85$, and the estimated bias is roughly $-363.85/23$.

Other examples of biased statistics include unadjusted R^2 in regression, scatterplot smoothing, regularized regression, and goodness of fit statistics when a variable selection procedure like stepwise regression is used.

2.1 Functional Statistics

There is a subtle point in the bootstrap bias estimate (equation 2)—it assumes that $\theta = \theta(F)$ and $\hat{\theta} = \theta(\hat{F})$ —in other words, that $\hat{\theta}$ is a *functional statistic* (Efron and Tibshirani (1993) also call this a *plug-in statistic*), that depends solely on the distribution, not on other factors such as sample size. For example, $\hat{\sigma}^2 = \text{Var}(\hat{F}_n)$ is functional, but $s^2 = 1/(n-1) \sum (x_i - \bar{x})^2 = n/(n-1) \text{Var}(\hat{F}_n)$ is not. Other non-functional statistics include adjusted R-squared in regression, stepwise regression, regularized regression, and scatterplot smoothing.

We can get odd results when bootstrapping non-functional statistics. For example, the printout above suggests that s^2 is biased. To the bootstrap, s^2 looks like $\hat{\theta} = s^2 = (23/22)\hat{\sigma}^2$, an estimate of $\theta = (23/22)\sigma^2$, and it estimates the bias accordingly.

Bootstrap SE estimates are not affected the same way as bias estimates, because they are calculated solely from the bootstrap statistics, whereas the bias estimate compares the bootstrap statistics to the observed statistic. Some confidence intervals are affected—bootstrap procedures typically provide confidence bounds for functional statistics.

3 Confidence Intervals

This section contains some additional information on confidence intervals, a combination of general material and discussion of two specific intervals—the expanded percentile interval, and tilting intervals.

3.1 Transformations

Table 1 gives rates of cardiovascular disease for subjects with high or low blood pressure. The high-blood pressure group was 2.12 times as likely to develop the disease.

Blood Pressure	Cardiovascular Disease
High	55/3338 = 0.0165
Low	21/2676 = 0.0078
Relative risk	2.12

Table 1: *Relative risk of cardiovascular disease.*

Figure 3 shows the bootstrap distribution for relative risk. The distribution is highly skewed, with a long right tail. Also shown is the bootstrap distribution for log relative risk; this distribution is less skewed. Both distributions exhibit bias; the summary statistics are:

	Observed	SE	Mean	Bias
Relative Risk	2.0996	0.6158	2.2066	0.1070
Log Relative Risk	0.7417	0.2625	0.7561	0.0143

Transformation invariance is desirable property for confidence intervals—if h is a monotone transformation and $\psi = h(\theta)$, an interval is transformation invariant if the endpoints of the confidence interval for ψ are $h(L)$ and $h(U)$, where (L, U) is the interval for θ .

The bootstrap percentile interval is transformation invariant. If one student does a confidence interval for $\theta =$ relative risk, and another for $\psi =$ log relative risk, they get equivalent answers; the percentile interval for relative risk is (1.31, 3.70), and for log-relative-risk is (0.273, 1.31) = (log(1.31), log(3.70)). In contrast, a t interval is not transformation invariant. The t interval with bootstrap SE for relative risk is $2.0996 \pm 1.96 \cdot 0.6185 = (0.893, 3.306)$; taking logs gives $(-0.113, 1.196)$. Those differ from t endpoints for log relative risk, $0.7417 \pm 1.96 \cdot 0.2625 = (0.227, 1.256)$.

For intervals that are not transformation invariant, the answer depends on the transformation. In this case, the choice of transformation determines whether the interval includes the parameter corresponding to no effect (ratio = 1, or log = 0). One hypothesis test based on these intervals would reject H_0 , the other would not.

Confidence intervals for population variance are another example of the effect of transformations. A t interval with bootstrap SE for σ^2 for the CLEC data (by bootstrapping s^2) is $(-174, 935)$; the left endpoint is nonsensical. A square-root transformation gives answers that make more sense; the corresponding interval for σ (bootstrapping s) is (4.25, 34.7), corresponding to an interval of (18.1, 1208) for σ^2 . Bootstrapping $\log(s^2)$ and transforming back gives yet another answer, an interval of (54.5, 2653) for σ^2 . In contrast, the bootstrap percentile interval gives compatible intervals using any transformation, (59.4, 931) for σ^2 .

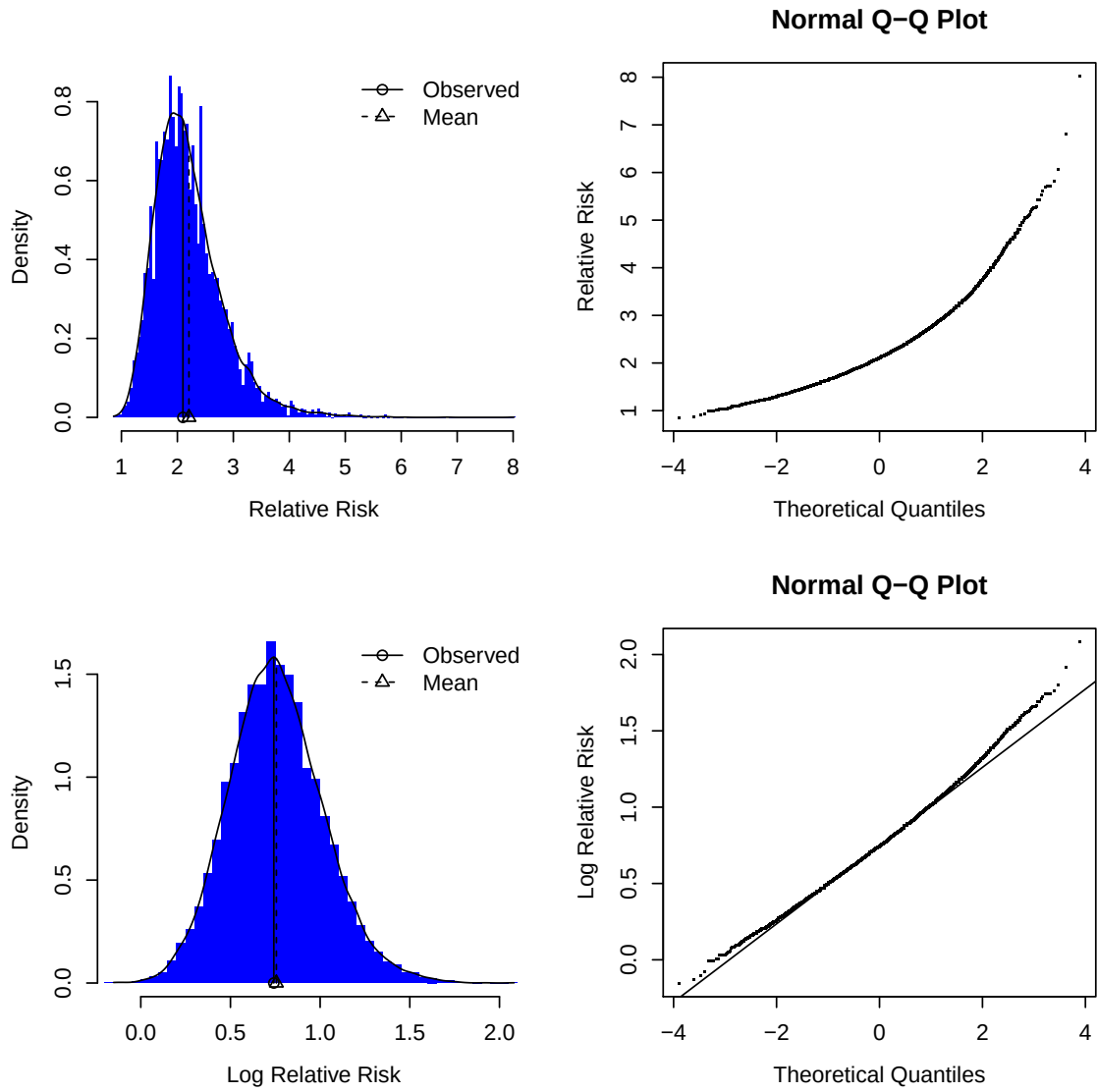


Figure 3: *Bootstrap relative risk.* The top panel shows the bootstrap distribution for relative risk. The bottom panel shows the bootstrap distribution for log relative risk.

3.2 Accuracy

A accurate confidence interval procedure includes the true value 95% of the time, and misses 2.5% of the time on each side. Different intervals, both formula and bootstrap, have trouble achieving this or even coming close, in different applications and with different sample sizes.

It is not correct for a 95% interval to miss 4% of the time on one side and 1% on the other—in practice almost all statistical findings are ultimately one-sided, so making an error on one side does not compensate for an error on the other. It would be rare indeed to find a report that says, “my new way of teaching was significantly different in effectiveness than the control” without also reporting the direction!

Suppose that the right endpoint of an interval is too low, so the interval misses 4% of the time on that side. I’d rather have an interval that is correct on the left side than one that is too low—because the combination of being too low on both sides gives an even more biased picture about the location of θ . A *biased confidence interval* has endpoints that are too low, or too high, on both sides.

First and Second Order Accurate: A hypothesis test or confidence interval is *first-order accurate* if the actual one-sided rejection probabilities or one-sided non-coverage probabilities differ from the nominal values by $O(n^{-1/2})$. They are *second-order accurate* if the differences are $O(n^{-1})$.

The usual t intervals and tests, percentile, and t interval with bootstrap standard errors are first-order accurate. The bootstrap t is second-order accurate. These statements assume certain regularity conditions, and apply to many common statistics, e.g. smooth functions of sample moments (for example, the correlation coefficient can be written as a function of $(\bar{x}, \bar{y}, \bar{x}^2, \bar{x}\bar{y}, \bar{y}^2)$), smooth functions of solutions to smooth estimating equations (including most maximum likelihood estimators), and generalized linear models. For details see (Efron and Tibshirani, 1993; Davison and Hinkley, 1997; DiCiccio and Romano, 1988; Hall, 1988, 1992).

There are many other bootstrap confidence intervals; in the early days of the bootstrap there was quite a cottage industry, developing second-order accurate or even higher order intervals. Some are described in (Efron and Tibshirani, 1993; Davison and Hinkley, 1997); for a review see DiCiccio and Efron (1996).

To be second-order accurate, a procedure needs to handle bias, skewness, and transformations. But just being second-order accurate isn’t enough in practice; an interval should also have good small-sample performance. A first-order accurate interval can be better in small samples than a second-order accurate interval, if it handles the “little things” better—things that have an $O(1/n)$ effect on coverage, with little effect for large n , but that matter for small n . The bootstrap percentile interval is poor in that regard. Next we consider a variation on the percentile interval that performs better.

n	$\sqrt{(n-1)/n}$	$z_{0.025}/t_{0.025,n-1}$	size	$\alpha'/2$
5	0.894	0.706	0.077	0.0010
10	0.949	0.866	0.048	0.0086
20	0.975	0.936	0.036	0.0159
40	0.987	0.969	0.030	0.0203
80	0.994	0.985	0.028	0.0226

Table 2: *Narrowness bias, and z/t , and adjusted quantiles.* Column 2 shows the narrowness bias. Column 3 shows the ratio between $z_{\alpha/2}$ and $t_{\alpha/2,n-1}$. Column 4 shows the combined effect of columns 2–3 on coverage—the actual one-sided non-coverage for normal populations for the interval $\bar{x} \pm z_{\alpha/2}\hat{\sigma}/\sqrt{n}$. Column 5 shows the nominal $\alpha'/2$ to use to correct for the two effects, see Section 3.3.

3.3 Expanded Percentile Interval

The bootstrap percentile interval performs poorly in small samples, because of the narrowness bias, and because it lacks a fudge factor to allow for variation in the standard error. The standard t interval handles both, using s in place of $\hat{\sigma}$ to avoid narrowness bias, and $t_{\alpha/2,n-1}$ in place of $z_{\alpha/2}$ as a fudge factor to allow for variation in s . We can interpret the t interval as multiplying the length of a reasonable interval, $\bar{x} \pm z_{\alpha/2}\hat{\sigma}$, by $a_{\alpha/2,n} = (t_{\alpha/2,n-1}/z_{\alpha/2})(s/\hat{\sigma})$, to provide better coverage. This multiplier is the inverse of the product of columns 2–3 of Table 2.

The fact that the t interval is exact for normal populations is a bit of a red herring—real populations are never exactly normal, and the multiplier isn’t correct for other populations. With positive kurtosis, for example, it would be better to use a t quantile with degrees of freedom lower than $(n-1)$, to allow for the greater uncertainty in s . Yet we continue to use the usual t interval, even though it is not perfect. Even for long-tailed distributions, where the fudge factor should be larger, using at least a partial fudge factor helps. (For binomial data we do use $z_{\alpha/2}$ instead of $t_{\alpha/2,n-1}$, because given p there is zero uncertainty in the variance.)

Similarly, we may take a sensible interval, the percentile interval, and adjust it to provide better coverage for normal populations; this adjustment will also help for most other populations. A simple adjustment is to expand both sides of a percentile interval by $a_{\alpha/2,n}$, but that would not be transformation invariant. We can achieve the same effect, while not losing transformation invariance, by adjusting the percentiles. If the bootstrap distribution is approximately normal then its quantiles are approximately

$$\hat{G}^{-1}(\alpha/2) \approx \hat{\theta} - z_{\alpha/2}\hat{\sigma}/\sqrt{n}.$$

We solve

$$\hat{\theta} - z_{\alpha'/2}\hat{\sigma}/\sqrt{n} = \hat{\theta} - t_{\alpha/2,n-1}s/\sqrt{n}$$

for α' ; this gives $z_{\alpha'/2} = \sqrt{n/(n-1)}t_{\alpha/2,n-1}$, or

$$\frac{\alpha'}{2} = \Phi\left(-\sqrt{\frac{n}{n-1}}t_{\alpha/2,n-1}\right).$$

Some values of $\alpha'/2$ are given in Table 2. For a nominal one-sided level of 0.025, the adjusted values range from 0.0010 at $n = 5$ to 0.0226 for $n = 80$.

Coverage using the adjusted levels is dramatically better, see Figures 20–22 in (Hesterberg, 2014b) and (Hesterberg, 1999a), though is still poor with $n = 5$.

This adjustment has no terms for bias or skewness; it only counteracts the narrowness bias and provides a fudge factor for uncertain width. Still, we see in Figures 21–22 of (Hesterberg, 2014b) that it also helps for skewness. For exponential populations, the expanded percentile is better than the t interval for $n \geq 7$; the unexpanded version is not better until $n \geq 35$.

The technique of using modified quantiles of the bootstrap distribution is motivated by the bootstrap BCa confidence interval (Efron, 1987), that uses modified quantiles to handle skewness and median bias. The BCa has no adjustment for narrowness or variation in SE, though these can be added. Expansion is the default for percentile intervals and BCa intervals in the *resample* package (Hesterberg, 2014a).

3.4 Smoothed Bootstrap

Here we present some additional material on the smoothed bootstrap, noting its value in percentile and t intervals.

The smoothed bootstrap is a compromise between parametric and nonparametric approaches; if we believe the population is continuous, we may sample from a continuous $\hat{F}$ rather than the empirical distribution $\hat{F}_n$. A convenient way to smooth is to sample from a *kernel density estimate*, e.g. from density $\hat{f}(v) = n^{-1} \sum_{i=1}^n \phi(x_i - v; h)$ where $\phi(\cdot; h)$ is the density for a normal distribution with mean 0 and standard deviation h . To generate a bootstrap sample from this distribution, we draw an ordinary bootstrap sample with replacement from the data, then add a random normal variate (with $\sigma = h$) independently to each observation.

Smoothing can produce negative values, even for data that must be positive, like time measurements. The middle column of Figure 6 has examples of smoothing producing negative values. A remedy is to transform the data (e.g. $\log(\text{time})$), smooth on that scale, then transform back.

Smoothing is not common, because it is more complicated, does not generalize well to multivariate and factor data, and it is rarely needed to make bootstrap distributions continuous. For statistics like the sample mean, if the original distribution was continuous then the number of distinct values in the theoretical bootstrap distribution is $\binom{2n-1}{n}$, so the distribution is practically continuous except for small n (Hall, 1986).

However, smoothing does have two particular advantages in bootstrapping. One is to correct for narrowness—the choice $h = s/\sqrt{n}$ corrects for the narrowness bias in the case of the sample mean (Hesterberg, 2004). However, it does not handle the t/z correction in the expanded percentile interval.

The second advantage is to adjust bootstrap t intervals for small samples. (Polansky, 2000) note that bootstrap t intervals for the mean of very small samples have high variability and long average length, because the denominator of the bootstrap t statistics can be very small. They find that using a smoothed bootstrap reduces the variability and average

length. The resulting interval is still second-order correct, but is more symmetrical than the usual bootstrap t interval, particularly for small samples.

3.5 Confidence Interval Pictures

Here are some pictures that show how confidence intervals should behave in different circumstances. In all cases the parameter is shown with a vertical line, the sampling distribution is on the top, and below that are bootstrap distributions. In the right side of Figure 4 and both sides of Figure 5, the confidence intervals for the samples that are second from the top and bottom should just touch the parameter, those in between should include the parameter, and the top and bottom ones should miss the parameter.

Figure 4 shows what happens in the nice case, of normally-distributed sampling distributions with no bias (and to make things simple, with known variance). Each bootstrap distribution is centered about the statistic for its sample. The bootstrap percentile interval and t -interval coincide, and each misses exactly the right fraction of the time on each side.

Simple Bias: The left side of Figure 5 shows what happens when there is simple bias, similar to that of unadjusted R-squared. The statistic is positively biased; the bootstrap distributions are similarly biased. The bias is $b = 0.2SD$ where $SD = \sqrt{\text{Var}(\hat{\theta})}$. A correct interval would be $\hat{\theta} - b \pm z_{\alpha/2}SE$, or $(\hat{\theta} - 1.84SE, \hat{\theta} + 1.44SE)$.

z intervals are symmetric about the corresponding statistic, so end up with one copy of the bias (from the bias in the original statistic). The intervals miss too often by being above θ , and not often enough below θ .

Bootstrap percentile intervals are even worse, because they get a second copy of the bias (the original bias, and bootstrap bias).

Skewness: The right side of Figure 5 shows what happens for unbiased statistics when the distribution is skewed; in this case, the mean of a gamma distribution with shape 9. The sampling distribution quantiles have roughly the same asymmetry ($\theta - 1.43SD, \theta + 1.81SD$) as in the bias example. The bootstrap distributions show the same asymmetry; the middle 90% (the 90% bootstrap percentile interval) is $(\hat{\theta} - 1.43SE, \hat{\theta} + 1.81SE)$.

A correct interval is $(\hat{\theta} - 1.13SE, \hat{\theta} + 2.75SE)$ (see the text beside the bold curves). A correct interval needs to reach many (short) standard errors to the right to avoid missing too often when $\hat{\theta} < \theta$ and standard errors are small.

This time the bootstrap percentile misses too often by being below θ , and not often enough by being above. Even though the interval is asymmetrical, it is not asymmetrical enough. The t interval is even worse.

Failure of intuition: The desired behavior for skewed data runs counter to our intuition. If we observe data with large observations on the right, our intuition may be to downweight those observations, and have the confidence interval reach farther left, because the sample mean may be much larger than the true mean. In fact, when the data show that the population is positively skewed, a good confidence interval must protect against

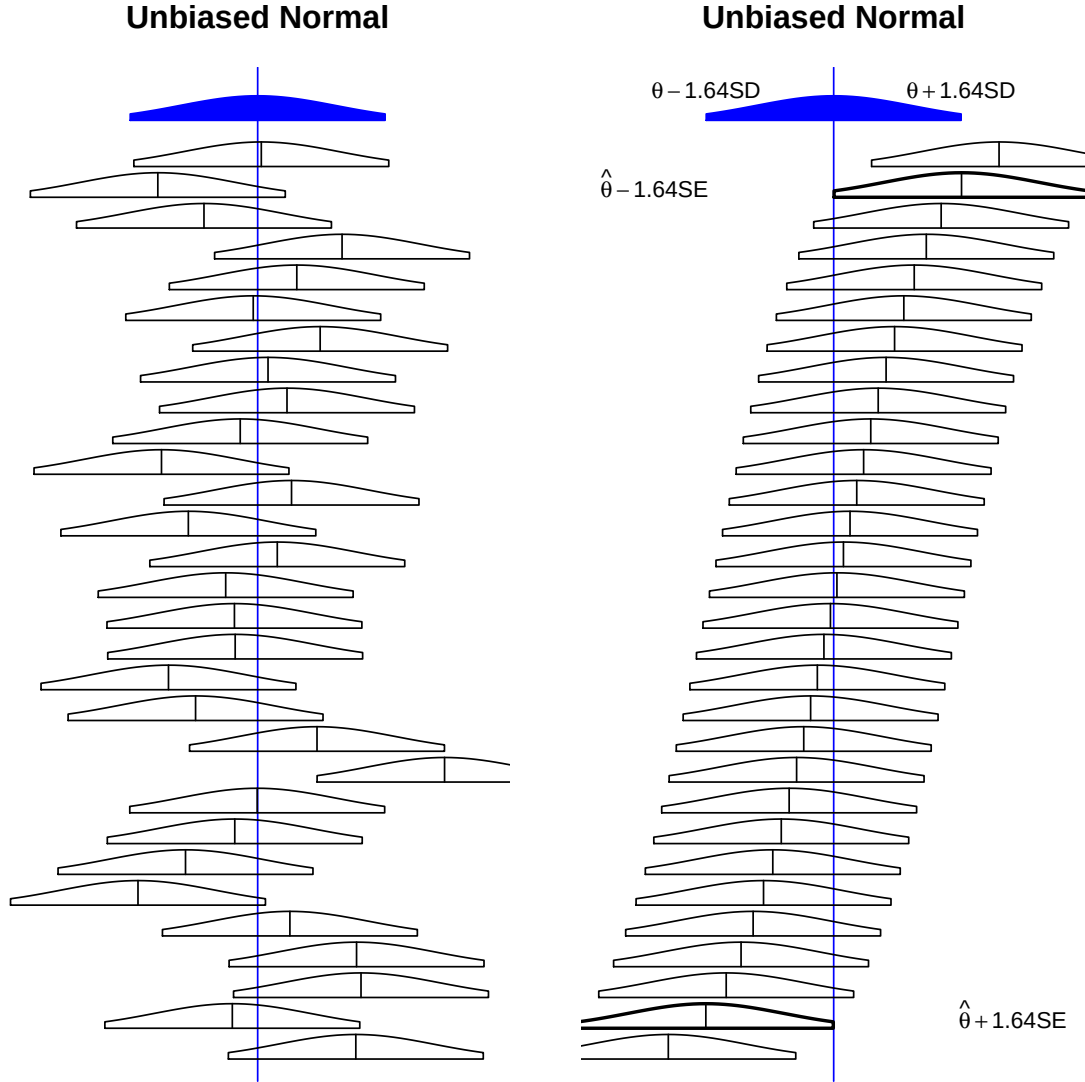


Figure 4: *Confidence intervals for normal with no bias.* The vertical lines correspond to true values of the parameter. The solid figures are the normally-distributed sampling distributions with no bias, truncated at the middle 90%. To have correct 90% coverage, a sample with $\hat{\theta}$ in that middle range should result in a confidence interval that includes θ , and others should miss θ . For simplicity, we assume that $SD^2 = \text{Var}(\hat{\theta}) = SE^2 = \text{Var}(\hat{\theta}^*)$. On the left are truncated bootstrap distributions, each for one random sample, centered at the corresponding $\hat{\theta}$. In this case, the bootstrap percentile interval and a z interval coincide, and both have the correct coverage; both CIs include θ when their $\hat{\theta}$ is in the middle 90% of the sampling distribution. On the right are bootstrap distributions, ordered by the $\hat{\theta}$, scaled so the bold distributions should just touch θ . A correct interval is $(\hat{\theta} - 1.64SE, \hat{\theta} + 1.64SE)$.

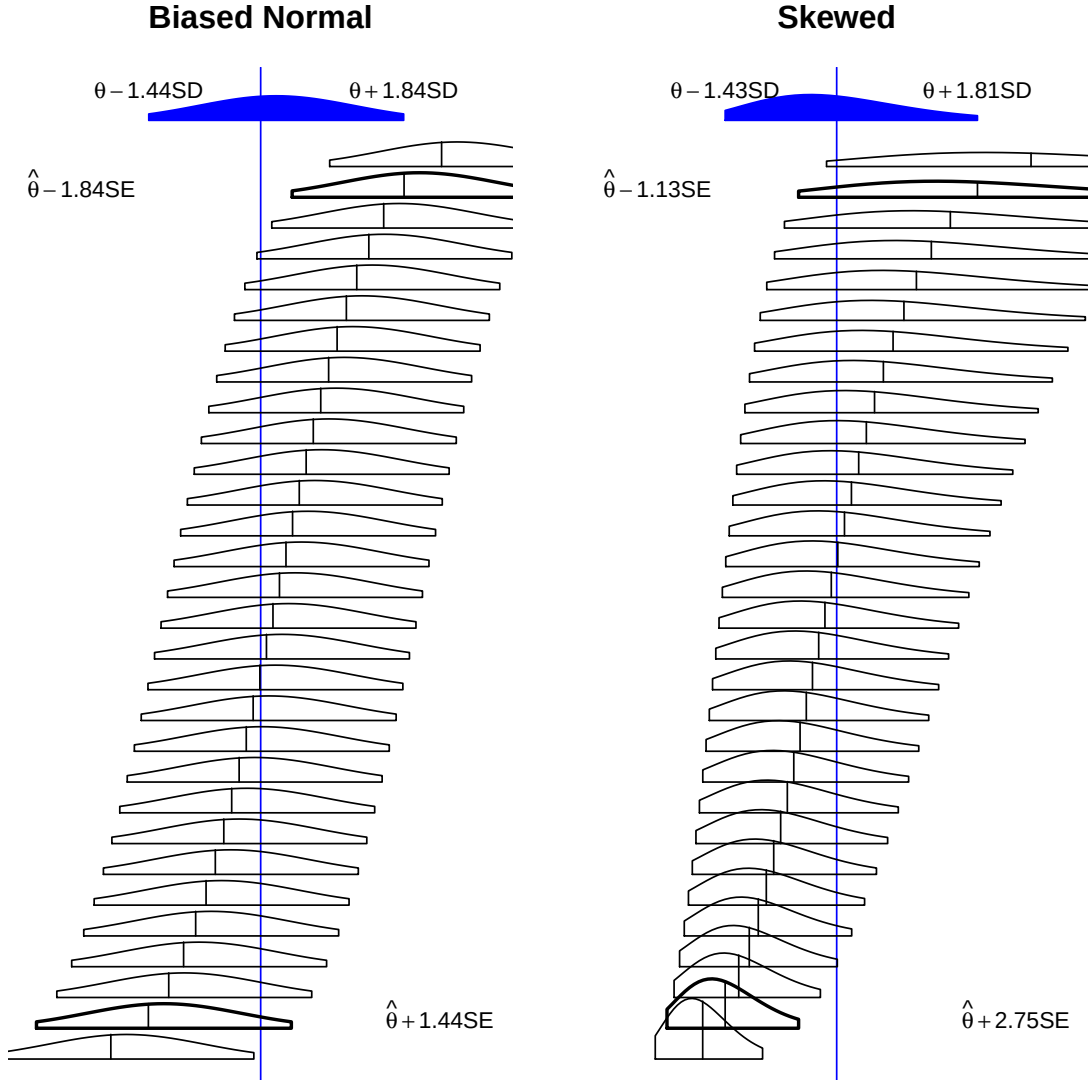


Figure 5: *Confidence intervals for bias, and acceleration.* The vertical lines correspond to true values of the parameter. The solid curves are sampling distributions, truncated at the middle 90%. On the left the sampling distribution, and bootstrap distributions, are normal with bias $0.2SD = 0.2SE$. For correct coverage, an interval should be $(\hat{\theta} - 1.84SE, \hat{\theta} + 1.44SE)$ (see the text beside the bold distributions). The bootstrap percentile interval is asymmetrical in the wrong direction: $(\hat{\theta} - 1.44SE, \hat{\theta} + 1.84SE)$. On the right the sampling distribution, and bootstrap distributions, are unbiased with skewness $2/3$ (the distributions are gamma with shape = 9). For correct coverage, an interval should be $(\hat{\theta} - 1.13SE, \hat{\theta} + 2.75SE)$. The bootstrap percentile interval $(\hat{\theta} - 1.43SE, \hat{\theta} + 1.81SE)$ is not asymmetrical enough. A t interval $(\hat{\theta} - 1.64SE, \hat{\theta} + 1.64SE)$ is even worse.

the possibility that we observed *fewer* than average observations from the right tail, and especially from the far right tail. If we're missing those observations, then $\bar{x}$ is too small, and s is also too small, so the interval must reach many standard errors to the right.

Conversely, we may have gotten more observations from the right tail than average, and the observed mean is too large—but in that case the standard error is inflated, so we don't need to reach so many standard errors to the left to reach the parameter.

Transformations: The 90% endpoints of the bootstrap distributions had roughly the same asymmetry in the bias and skewness examples: $(-1.44, 1.84)$ vs $(-1.43, 1.81)$. We could get the same asymmetry by applying a nonlinear transformation $\psi = \exp(0.15\mu)$ to the case of normal with no bias, with $\hat{\psi} = \exp(0.15\bar{x})$. This transformation gives sampling distributions and bootstrap distributions with the same asymmetry as the bias example, $0.28/0.22 \approx 1.84/1.44$. In this case a bootstrap percentile interval would be correct, and a t interval would not.

Need more information: We can't tell just from the asymmetry of the endpoints whether a correct interval should be asymmetrical to the right or left. The correct behavior depends on whether the asymmetry is caused by bias, skewness, transformations, or a combination thereof. We need more information—and second-order accurate bootstrap confidence interval procedures collect and use such information, explicitly or implicitly.

But lacking that information, the percentile interval is a good compromise, with transformation invariance and a partial skewness correction.

3.6 Tilting

Bootstrap tilting (Efron, 1981) is an approach to confidence intervals and hypothesis tests that I believe has great pedagogical potential for Mathematical Statistics; it nicely connects parametric and nonparametric statistics, and can help students understand the relationship between parameters and sampling distributions, and better understand confidence intervals. Empirical likelihood (Owen, 2001) is related.

The initial exposition (Efron, 1981) is for confidence intervals, but the method is easier to understand starting with hypothesis tests. Consider testing $H_0 : \mu = \mu_0$ is n i.i.d. observations from a population F . We begin with a visual explanation, followed by a more formal explanation. Figure 6 shows the process. For any value of $\mu = \mu_0$ (each of the first five rows uses a different value of μ as an example), we find probabilities p_i such that $\sum_{i=1}^n p_i x_i = \mu_0$. The corresponding weighted empirical distribution has mean μ_0 . When $\mu_0 = \bar{x}$ as in row 3, $p_i = 1/n \forall i$. When $\mu_0 < \bar{x}$ as in rows 1–2, p_i is larger for small values of x ; conversely when $\mu_0 > \bar{x}$ as in rows 4–5. Next, we draw bootstrap samples with replacement from the weighted empirical distributions, drawing x_i with probability p_i . The bootstrap distributions are shown in the right column of the figure. Each distribution has mean μ_0 , and is a null distribution for the corresponding hypothesis test. The one-sided P -value is the fraction of the bootstrap distribution that exceeds the observed $\bar{x}$; e.g. for the top row the one-sided P -value for $H_a : \mu > \mu_0$ is the area under the bootstrap density curve to the right of the gray vertical line at $\bar{x}$.

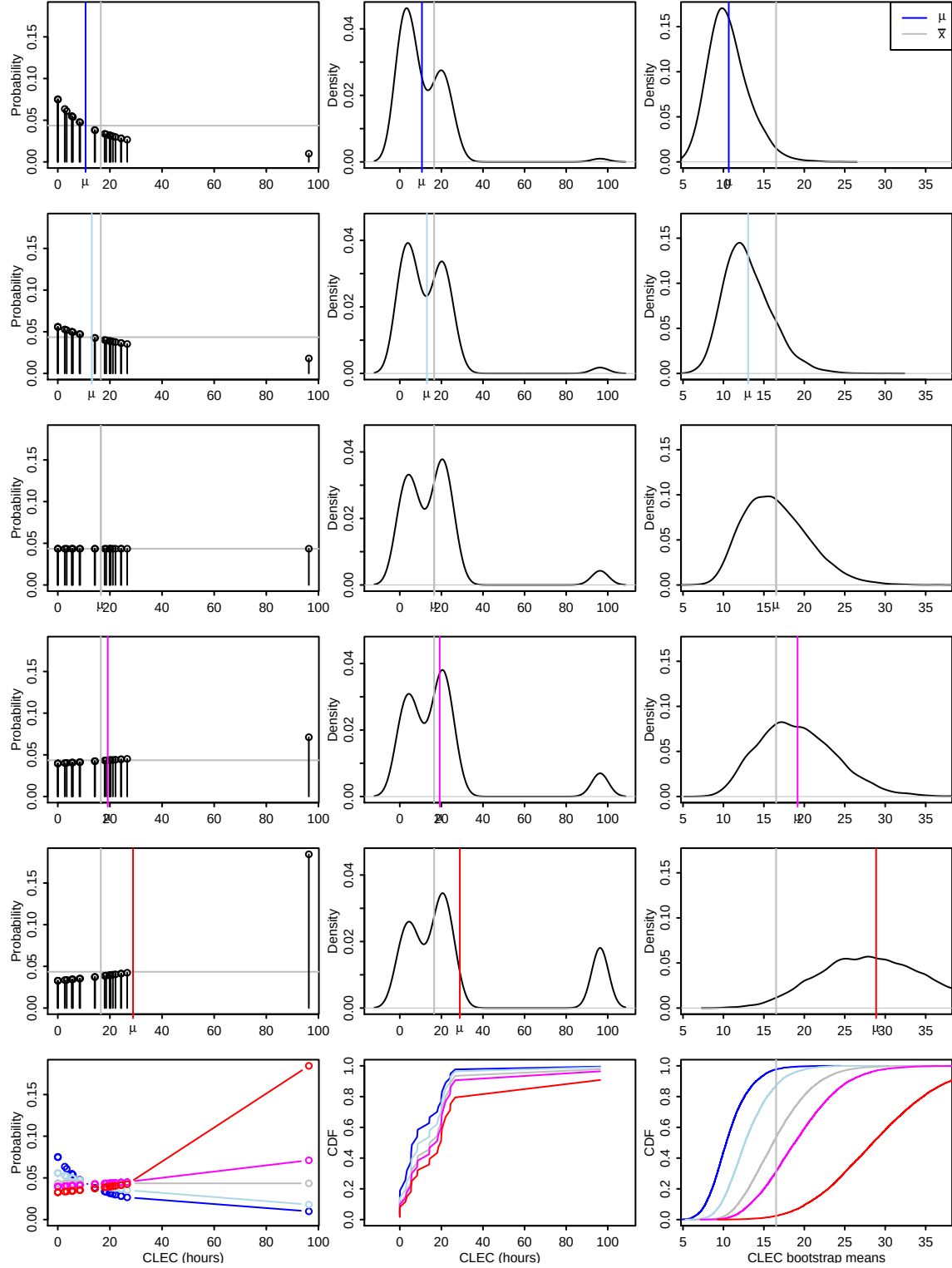


Figure 6: *Bootstrap tilting, for the CLEC mean.* The top five rows correspond to different values of μ . The left column contains probability vectors such that $\sum p_i x_i = \mu$. The middle column shows the corresponding weighted density estimates; the bottom are CDFs for those weighted distributions (lines connect the midpoints of the vertical jumps for each CDF). The right column shows bootstrap distributions, when sampling from the CLEC data with probabilities p_i .

Tilting turns the data into a one-parameter parametric family, with parameter μ . The left and middle columns correspond to views of the population $\hat{F}$ indexed by μ , and the right column to the sampling distribution for $\bar{x}$ from $\hat{F}$. For skewed data the parametric family preserves the appropriate mean-variance relationship; for larger values of μ , both $\hat{F}$ and the sampling distribution have larger variance.

Tilting confidence intervals consist of those values of θ that have $\hat{\theta}$ in the middle 95% of their (estimated) sampling distributions. Since there is a 95% chance that $\hat{\theta}$ is in the middle 95% of the (true) sampling distribution, there is a 95% chance that (the true) θ is included in the interval.

The endpoints of the interval are the values of μ for which the one-sided P -values are $\alpha/2$; those are the interval endpoints. In this case the first and fifth rows have one-sided P -values of 0.025; the 95% confidence interval is (10.7, 28.9). For comparison, the bootstrap t interval is (10.6, 31.0), percentile interval is (10.2, 25.4), and t with bootstrap SE is (8.3, 24.7). The tilting and bootstrap t intervals agree most closely, though the tilting interval is narrower (tilting intervals may suffer from some of the same small-sample issues as the percentile interval).

These pictures might help students better understand confidence intervals. In creating confidence intervals, we need to think backward—determining what values of θ could reasonably have given rise to the observed $\hat{\theta}$. This is like tossing a ball on a hill; we observe where the ball landed ($\hat{\theta}$), and need to infer where it was thrown from (θ). We need to determine what values of θ have $\hat{\theta}$ in the middle 95% of their sampling distribution. In contrast, the bootstrap percentile interval reverses the roles of $\hat{\theta}$ and θ , and symmetric t intervals obfuscate the roles.

Figure 7 shows the same process, for the much larger ILEC data. As with the ILEC figure, the two extreme values of μ correspond to a 95% confidence interval, (7.76, 9.19). This is a smaller range of μ , and over this range the differences between the populations appear much smaller, particularly in the density and CDF plots in the middle column. Still, with the larger sample, those subtle differences in the population are enough to give striking differences in the sampling distributions. Also, while the mean-variance relationship is still present, the narrow range of μ does not result in large differences in variance. The biggest lesson here is that for large n , relatively small differences in the population can have relatively large effects on the sampling distribution.

Now, we give a more formal description of tilting. We make the working assumption that F has support only on the observations; then F is determined by weights (probabilities) on the observations, $(p_1, \dots, p_n)$, with $p_i \geq 0$ and $\sum p_i = 1$. This transforms the original nonparametric problem into a parametric application with n parameters in an $(n - 1)$ -dimensional space. The mean for this distribution is

$$\mu = \sum p_i x_i. \quad (3)$$

The likelihood of the observed data is $\prod p_i$; the maximum likelihood estimate is $\hat{\mathbf{p}} = (1/n, \dots, 1/n)$. The MLE subject to the null hypothesis is the vector that maximizes $\prod p_i$ subject to $p_i \geq 0$, $\sum p_i = 1$, and $\sum p_i x_i = \mu_0$. The solution, obtained using calculus of

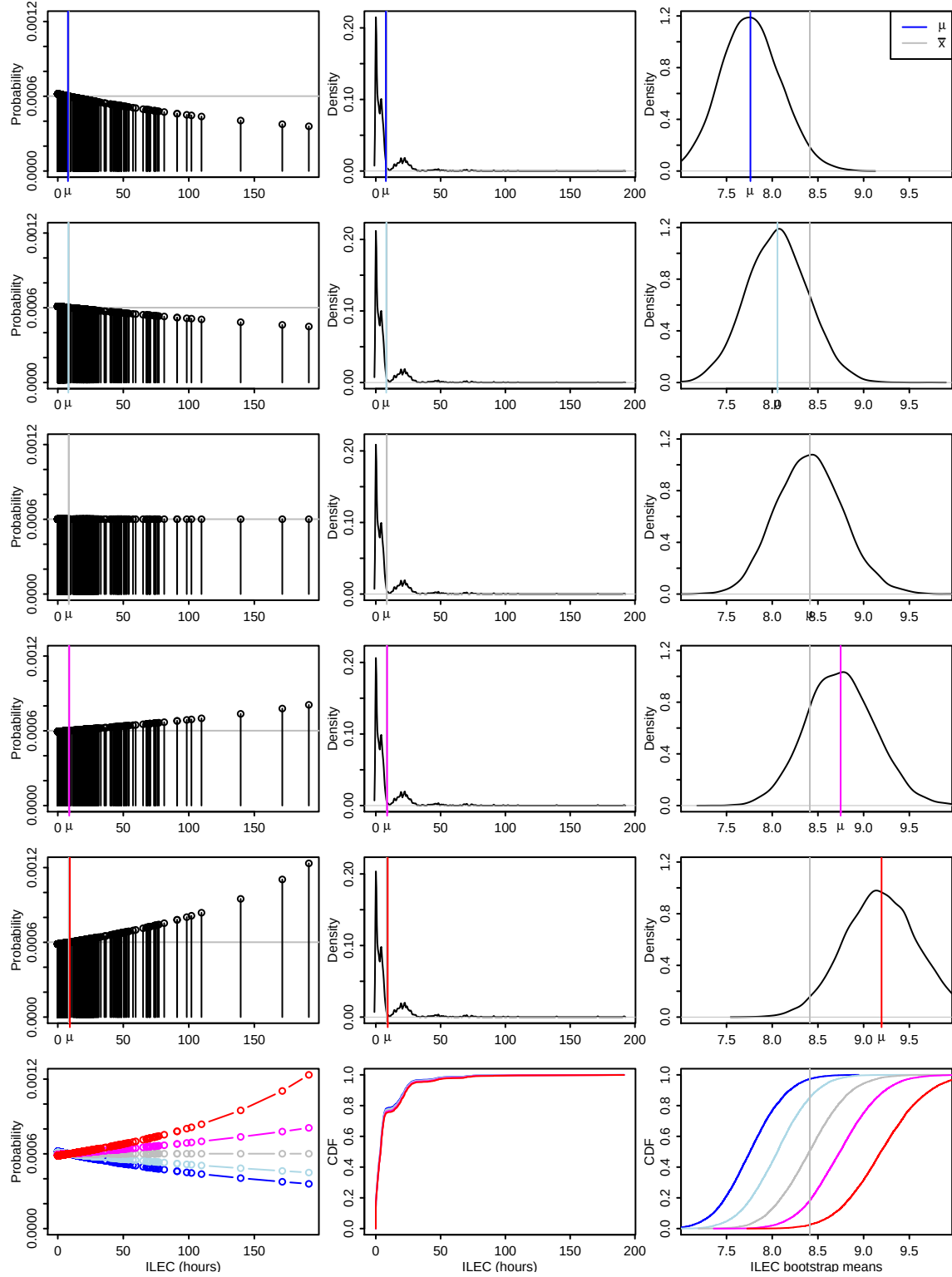


Figure 7: *Bootstrap tilting, for the ILEC mean.* The top five rows correspond to different values of μ . The left column contains probability vectors such that $\sum p_i x_i = \mu$. The middle column shows the corresponding weighted density estimates; the bottom are CDFs for those weighted distributions (lines connect the midpoints of the vertical jumps for each CDF). The right column shows bootstrap distributions, when sampling from the ILEC data with probabilities p_i .

variations, can be written in the form

$$p_i = \frac{c(\tau)}{1 - \tau(x_i - \bar{x})} \quad (4)$$

where τ is a *tilting parameter* and $c(\tau)$ normalizes the weights to sum to 1. This solution reduces the problem to a one-parameter parametric family, indexed by τ (or equivalently by μ). τ is bounded, $1/(\min(x_i) - \bar{x}) < 1/(\max(x_i) - \bar{x})$, so that the denominator is strictly positive.

Note how different values of τ tilt the probabilities, in the left column of the figures.

To test $H_0 : \mu = \mu_0$, we find the value of τ that gives $\mu = \mu_0$, draw bootstrap samples with replacement from the resulting probabilities p_i , i.e. from the weighted empirical distribution, and compute the P -value as the fraction of the bootstrap statistic values that are at least as extreme as the observed mean.

Another variation of the method uses *exponential tilting*, of the form $p_i = c(\tau) \exp(\tau x_i)$. In either case, there is a 1-1 correspondence between τ and $\mu = \sum p_i x_i$. Both variations are *least-favorable families* within the n -parameter space; restricting inference to these one-parameter families yields asymptotically-equivalent inferences, though the exponential-tilting versions are narrower and undercover more badly in small samples.

For other statistics, we replace the weighted mean (equation 3) with the corresponding θ computed for the weighted distribution. There is not generally a simple formula for the probabilities.

For more information about tilting, see (Efron, 1981; Davison and Hinkley, 1997; Hesterberg, 1999b,a); DiCiccio and Romano (1990) show that tilting intervals are second-order accurate. The intervals are transformation-invariant.

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